

A Meta-Unified Global Cyber Threat Intelligence Model for Industrial Cross-Cloud Networks

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Abstract—Industrial cross-cloud networks (ICCNs) combine services from multiple providers to support critical operations, but this diversity also creates major security challenges. Because each provider follows different protocols and safeguards, responses to cyber threats are often inconsistent and delayed, reducing the effectiveness of existing Cyber Threat Intelligence (CTI) systems. To address this gap, we propose a three-layered CTI framework that unifies learning and adaptation across heterogeneous clouds. The first layer implements a two-tier federated learning (FL) strategy, developing two global models: *MLP-SimiFed* and *MLP-Non-SimiFed*, within each industry cloud to manage scalability and heterogeneity. The second layer aggregates these models across distributed clouds into a Unified Global Model, strengthening collaborative defense. The third layer leverages transfer learning to produce the *Meta Unified Global Cyber Threat Intelligence* (MUG-CTI) model, which enables swift adaptation to emerging threats. Through test-bed simulation, MUG-CTI demonstrates superior threat management, achieving up to 20.7% and 43.06% higher accuracy, and reducing hamming loss by up to 73.8% and 98.3% compared to Federated Learning and SVM-driven methods, respectively.

Index Terms—Cross-clouds, cyber threat intelligence, cybersecurity, industry clouds, multi-dimensional federated learning.

I. INTRODUCTION

THE rapid expansion of cloud computing across diverse industries has given rise to complex cross-cloud environments, where data and applications are distributed across multiple platforms [1]. This trend has significantly expanded the attack surface, making Industrial Cross-Cloud Networks (ICCNs) particularly vulnerable to sophisticated cyber threats. According to a 2024 report by International Business Machines (IBM) Corporation Security, the global average cost of a data breach has increased to \$4.67 million, with 64% of breaches involving data stored in multi-cloud environments [2]. Similarly, the “2024 Cyber Threats Report” estimates that cybercrime will cost the world \$10.5 trillion annually by 2025, with cross-cloud networks identified as prime targets due

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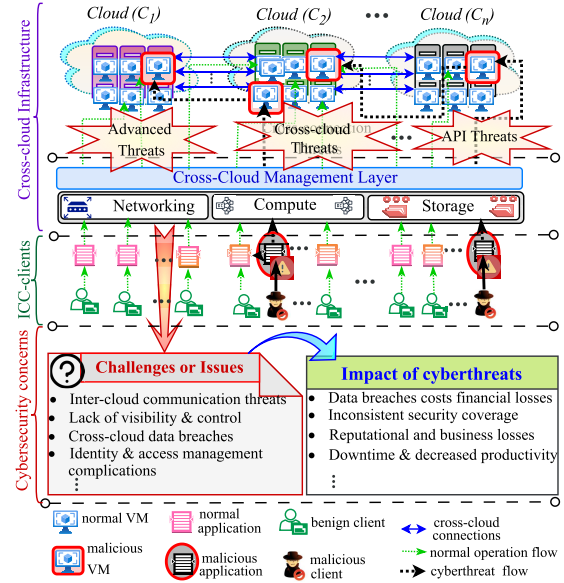


Fig. 1. Industry cross-cloud networks and cybersecurity challenges.

to their complexity and interconnectivity [3]. Furthermore, a Gartner survey forecasts that by 2026, 85% of organizations will adopt a cloud-first strategy, further underscoring the urgency of robust security frameworks [4].

As illustrated in Fig. 1, ICCNs integrate and manage cloud services from multiple providers through a cross-cloud management layer to improve flexibility, scalability, and resilience. This integration optimizes the use of networking, compute and storage resources while ensuring interoperability among heterogeneous platforms. Industry Cloud Clients (ICCs) operate collaboratively within these environments, but malicious actors may exploit vulnerabilities in inter-cloud communication and shared infrastructures. Maintaining security across such interconnected systems is inherently challenging due to the coexistence of diverse provider protocols and inconsistent defense mechanisms [5], [6]. These conditions often result in fragmented threat detection, delayed responses, and identity management challenges. Advanced persistent threats (APTs), ransomware, and other sophisticated attacks exploit these gaps, leading to financial losses, reputational damage, and productivity decline [7], [8], [9]. Addressing these issues requires more than traditional intrusion detection or siloed cloud security measures. A comprehensive CTI capability is needed to unify detection, analysis and mitigation in heterogeneous clouds. CTI is uniquely positioned to transform raw

distributed threat signals into actionable intelligence, enabling proactive and real-time defense strategies. In the context of ICCNs, CTI must be able to (i) capture the diversity of threat behaviors across multiple clouds, (ii) reconcile heterogeneous and dynamic infrastructures, and (iii) adapt rapidly to evolving attack vectors while minimizing delays in response. Without such intelligence-driven approaches, ICCNs remain highly susceptible to fragmented defenses that cannot match the sophistication of modern adversaries.

A. Related Work

Recent advancements in CTI have introduced several models to secure industrial cloud environments. Yazdinejad et al. [10] employ federated learning for privacy-preserving, cluster-based anomaly detection in Industrial Internet of Things (IIoT), achieving high accuracy with minimal bandwidth. Li et al. [11] combine federated learning with fog/edge computing to detect and mitigate zombie nodes in IIoT networks, improving accuracy, reducing mitigation time, and increasing the attack cost for adversaries. Liu et al. [12] propose a secure and lightweight feature selection system (SeiFS) for machine learning (ML) tasks, enabling efficient interactions between multiple entities and a single server. Hodayoun et al. [13] introduce deep ransomware threat hunting and intelligence system (DRTHIS) using long short-term memory (LSTM) and convolutional neural network (CNN), achieving high accuracy in identifying known and unseen ransomware families in fog layer nodes. Mm et al. [14] propose a support vector machine (SVM)-based distributed denial-of-service (DDoS) detection algorithm with four phases: training, pre-processing, feature selection, and classification. Gao et al. [15] present heterogeneous information network-based cyber threat intelligence (HinCTI), a system using heterogeneous networks and graph convolutional networks (GCN) to model CTI and identify threat types based on infrastructure node similarities. Hu et al. [16] propose Graph2vec+RF, which uses graph embedding and Random Forest to enable early and accurate network intrusion detection from initial network packets. Sundaramoorthy et al. [17] propose a hybrid intrusion detection system (IDS) for cloud computing and wireless sensor networks (WSN)s using multi-layer perceptron (MLP), optimizing detection accuracy and reducing false positives. A data-driven decision support framework is designed in [18] to optimize cybersecurity forensic investigations by using a repository of known adversarial tactics and threat intelligence to guide investigators in selecting the most efficient next steps, thereby, reducing the duration and cost of the investigation.

B. Research Gaps and Our Contributions

Despite significant advances in CTI research, several challenges persist in securing Industrial Cross-Cloud Networks (ICCNs). Existing CTI and Federated Learning (FL) approaches often address only a single dimension, such as non-independent and non-identically distributed (non-IID) data handling or privacy preservation, but lack holistic mechanisms to simultaneously manage dynamic topologies, model stability, and evolving threats. Consequently, threat responses are often

fragmented and delayed, limiting the capability of current systems to provide unified, real-time protection in ICCNs. The key limitations of existing research can be summarized as follows:

- 1) *Model-level adaptation*: FL in ICCNs must handle highly non-IID data and heterogeneous local infrastructures, which can lead to biased or unstable global models [10], [11].
- 2) *Threat-evolution adaptation*: ICCNs encounter continuously emerging and polymorphic attacks, including supply-chain compromises and AI-driven malware, requiring rapid incorporation of new threat intelligence into existing CTI models [15], [16].
- 3) *Coverage of diverse threats*: Many existing systems, such as DRTHIS [13], DDoS detection methods [14], and hybrid IDSs [17], focus on specific attack types and fail to address the full spectrum of multi-vector threats prevalent in ICCNs.
- 4) *Handling heterogeneity and dynamism*: CTI and FL methods often struggle to adapt to the heterogeneous and dynamically changing infrastructure of ICCNs [10], [11], [15], [18].
- 5) *Real-time adaptability*: Current approaches generally lack the capability to continuously update and respond to rapidly evolving threats across distributed cloud infrastructures [10], [11].

To address the identified gaps, we introduce a multi-diverse CTI model, conceptualized as a three-layered Meta-Unified Global CTI (MUG-CTI) framework, as depicted in Fig. 2. The first and second layers employ Multi-Layer Perceptron (MLP)-based heterogeneous Federated Learning (FL) strategies, both similarity-driven and non-similarity-driven for scalable, privacy-preserving collaboration within and across industry clouds. The third layer integrates Transfer Learning to ensure real-time adaptability and knowledge transfer. Collectively, this layered design provides a synchronized, proactive, and globally unified defense mechanism configured for industrial cross-cloud networks. The primary contributions of this work are as follows:

- 1) *Novel Meta-Unified CTI Architecture*: We propose a tri-layered Meta Unified Global Cyber Threat Intelligence (MUG-CTI) model, representing a first-of-its-kind architecture that unifies threat detection, analysis, and mitigation across heterogeneous industrial cross-cloud networks. This design addresses the limitations of fragmented and delayed threat responses in existing systems.
- 2) *Layer-wise Heterogeneous Federated Learning Strategy*: The first layer introduces a novel two-tier FL optimization that develops two distinct global models: MLP-SimiFed and MLP-Non-SimiFed, within each industry cloud. This approach is unique in explicitly handling non-IID data distributions and heterogeneous infrastructures, providing stable and scalable local-global model learning.
- 3) *Unified Global CTI Aggregation Mechanism*: The second layer presents a novel aggregation strategy that combines multiple global models from distributed clouds into a single Unified Global Model. This ensures a

TABLE I
NOTATIONS WITH THEIR DESCRIPTION

Notation: Description	Notation: Description
IC : Industry Cloud	Z : Number of ICs
K : Number of cyber threat databases	$\mathcal{D}_k$: k^{th} local dataset
N : Number of samples in $\mathcal{D}_k$	$\ell(\cdot, \cdot)$: cross-entropy loss function
W_l : Weight matrix for layer l	$\mathcal{B}_l$: Bias vector for layer l
$\sigma(\cdot)$: ReLU activation function	$\hat{\mathcal{Y}}$: Prediction output of model
η : Learning rate	$\nabla_{\theta} \mathcal{L}(\theta)$: Gradient of the loss function
$\mathcal{G}_S, \mathcal{G}_{NS}$: Similar, Non-similar groups	l : Number of layers
α_i, β_i : Weights assigned to Similar and Non-Similar models	m : Number of layers
z : Number of communication rounds	b : Bias vector
θ_k : k^{th} MLP model	θ_{SGM} : MLP-SimiFed global model
θ_{N-SGM} : MLP-Non-SimiFed global model	θ_{UG-CTI} : Unified global CTI model
μ : learning rate for model updates	θ_{newNS} : Newly aggregated model

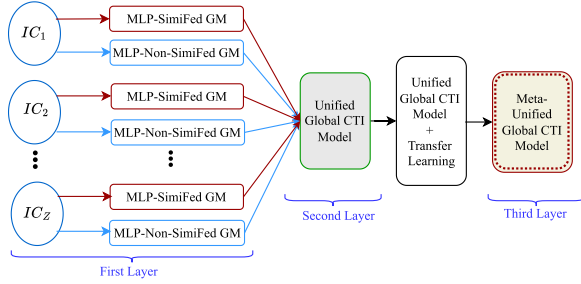


Fig. 2. Abstract view of MUG-CTI model.

consistent and robust security posture across all cloud platforms, which is not addressed in existing FL- or CTI-based approaches.

- 4) *Adaptive Transfer Learning for Real-Time Threat Intelligence*: The third layer leverages Transfer Learning to generate the Meta Unified Global CTI (MUG-CTI) model, enabling rapid adaptation to emerging and polymorphic threats. This layer represents a novel contribution by integrating adaptive intelligence for real-time threat evolution in ICCNs.

C. Paper Organization

Section II presents the tri-layered proposed CTI model describing *First Layer: MLP-SimiFed and MLP-Non-SimiFed Global Models* (Section II-A); *Second Layer: Unified Global CTI Model* (Section II-B); *Third Layer: Meta Unified Global CTI Model* (Section II-C); and operational design and complexity analysis (Section II-D). The detailed description of the performance evaluation and comparison is discussed in Section III. Section IV summarizes the conclusions and outlines future research directions. Table I summarizes the notations and mathematical symbols used throughout the paper along with their descriptions.

II. MUG-CTI MODEL

Consider a cross-cloud environment with Z geographically distributed industry clouds $\{IC_1, IC_2, \dots, IC_Z\}$, each cloud maintains K cyber threat repositories $\{CTB_1^{IC_1}, CTB_2^{IC_1}, \dots, CTB_K^{IC_1}\}$, where $K \geq 1$ and $i \in \{1, 2, \dots, Z\}$. These repositories are used to develop a comprehensive and diverse CTI model for global cybersecurity in the Industrial Cross-Cloud Network (ICCN). As illustrated in Fig. 3, the proposed multi-diverse CTI model is a three-layered meta-unified global framework that leverages Multi-Layer

Perceptron (MLP)-based, two-tier heterogeneous Federated Learning (FL) optimization techniques for real-time learning and model adaptation.

In the *first layer*, two global models are developed at each industry cloud platform using first-level heterogeneous FL optimization: the *MLP-SimiFed Global Model* (SGM) and the *MLP-Non-SimiFed Global Model* (N-SGM). The *second layer* employs second-level heterogeneous FL optimization to build a *Unified Global CTI* (UGM-CTI) model by aggregating multiple MLP-SimiFed and MLP-Non-SimiFed global models (developed in the first layer) in geographically distributed industry clouds $\{IC_1, IC_2, \dots, IC_Z\}$. In the *third layer*, the diversified cyber threat intelligence from the Unified Global CTI model is further enhanced using *Transfer Learning* to learn new and advanced cyber threats, culminating in the development of the *Meta Unified Global Cyber Threat Intelligence* (MUG-CTI) model. The operational workflow for each of the three layers is detailed in the following subsections.

A. First Layer: MLP-SimiFed and MLP-Non-SimiFed Global Models

1) *Local Model Training*: For each k^{th} cyber threat database (CTB_k) such that $k = 1, 2, \dots, K$ associated with respective cluster of servers at i^{th} industry cloud platform IC_i where $i = 1, 2, \dots, Z$, a local MLP model $f(\cdot; \theta_k)$ is trained on the local dataset $\mathcal{D}_k$. The objective function for local model training is defined in Eq. (1):

$$\mathcal{L}_k(\theta_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} \ell(f(\mathcal{S}_{i,k}; \theta_k), \mathcal{Y}_{i,k}) + \frac{\lambda}{2} \|\theta_k\|^2 \quad (1)$$

where θ_k represents the parameters of the MLP model at IC_k , $\mathcal{D}_k = \{(\mathcal{S}_{i,k}, \mathcal{Y}_{i,k})\}_{i=1}^{N_k}$ is the local dataset, N_k is the number of samples in $\mathcal{D}_k$, $\ell(\cdot, \cdot)$ denotes the cross-entropy loss function, and λ is the regularization parameter.

The MLP model consists of multiple layers, each computing a weighted sum followed by an activation function depicted in Eq. (2):

$$\mathcal{Z}_l = \sigma(\mathcal{W}_l \mathcal{Z}_{l-1} + \mathcal{B}_l) \quad (2)$$

where $\mathcal{Z}_0 = \mathcal{S}$ represents the input vector, $\mathcal{W}_l$ denotes the weight matrix for layer l , $\mathcal{B}_l$ is the bias vector for layer l , and $\sigma(\cdot)$ is the ReLU activation function.

The output of the final layer, providing the prediction output of the model $\hat{\mathcal{Y}}$, is given by Eq. (3):

$$\hat{\mathcal{Y}} = f(\mathcal{S}; \theta) = \mathcal{W}_L \mathcal{Z}_{L-1} + \mathcal{B}_L \quad (3)$$

The discrepancy between the predicted outputs $\hat{\mathcal{Y}}$ and the true labels $\mathcal{Y}$ is measured using a loss function. For classification tasks, cross-entropy loss is used (Eq. 4):

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(\hat{\mathcal{Y}}_i, \mathcal{Y}_i) \quad (4)$$

The parameters θ are updated using Gradient Descent using Eq. (5) while Stochastic Gradient Descent (SGD) is used for large datasets as mentioned in Eq. (6):

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}(\theta^{(t)}) \quad (5)$$

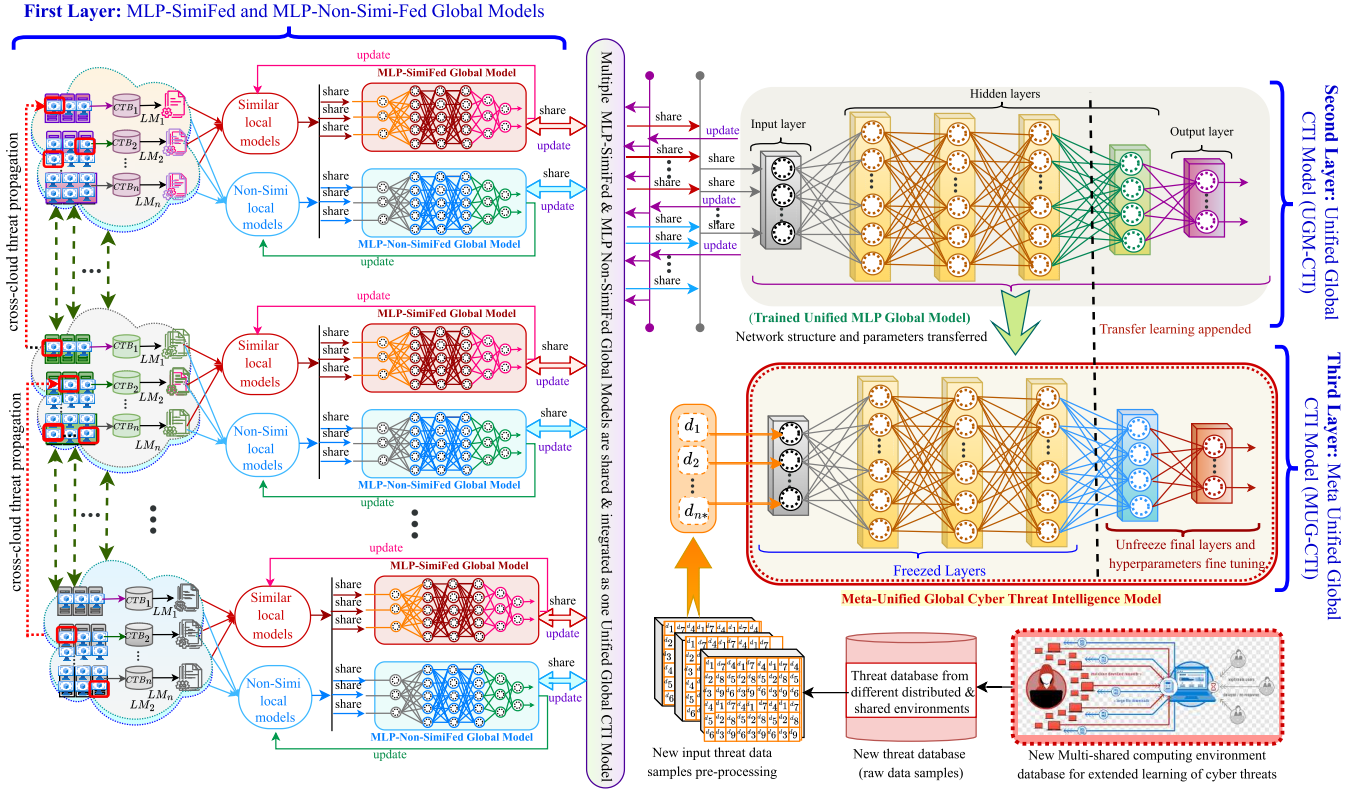


Fig. 3. Schematic workflow of meta-unified global cyber threat intelligence (MUG-CTI) model.

where η represents the learning rate, and $\nabla_{\theta}\mathcal{L}(\theta)$ is the gradient of the loss function.

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}_{\text{mini-batch}}(\theta^{(t)}) \quad (6)$$

To prevent overfitting, regularization techniques such as L2 regularization are applied using Eq. (7):

$$\mathcal{L}_{\text{reg}}(\theta) = \mathcal{L}(\theta) + \frac{\lambda}{2} \|\theta\|^2 \quad (7)$$

2) Building SGM and N-SGM Models:

a) *Model similarity computation:* The similarity between local models θ_i and θ_j is computed using Cosine Similarity, as shown in Eq. (8). Cosine similarity is selected because it is scale-invariant and focuses on the orientation of parameter vectors, making it well-suited when clients train with varying batch sizes or parameter magnitudes.

$$\text{Sim}(\theta_i, \theta_j) = \frac{\theta_i \cdot \theta_j}{\|\theta_i\| \|\theta_j\|} \quad (8)$$

b) *Grouping models:* Based on the similarity scores, models are classified into Similar ($\mathcal{G}_S$) and Non-Similar ($\mathcal{G}_{NS}$) groups as mentioned in Eqs. (9) and (10). The similarity threshold (ϵ) was set equal to 0.6 (chosen through validation to balance group sizes and reduce noisy merges, we observed useful ranges (0.5-0.7)).

$$\mathcal{G}_S = \{\theta_i \mid \text{Sim}(\theta_i, \theta_j) \geq \epsilon\} \quad (9)$$

$$\mathcal{G}_{NS} = \{\theta_i \mid \text{Sim}(\theta_i, \theta_j) < \epsilon\} \quad (10)$$

where ϵ is a predefined threshold that determines the grouping of models. This approach allows for diversified model

aggregation in Federated Learning scenarios, where similar models enhance generalization, while non-similar models retain specialized knowledge. The grouping of local models into Similar ($\mathcal{G}_S$) and Non-Similar ($\mathcal{G}_{NS}$) groups depends on the manually set cosine similarity threshold ϵ . A higher ϵ leads to stricter grouping, improving specialization but reducing generalization, while a lower ϵ increases generalization but risks merging dissimilar models. Empirical evaluation shows that $\epsilon = 0.6$ provides a balanced trade-off, ensuring stable grouping and improved model performance in terms of cross-entropy and Hamming loss.

3) First-Level Heterogeneous FL Optimization:

a) *Similarity grouping:* As illustrated in Fig. 3, the similarity between local models θ_i and θ_j is calculated using cosine similarity as given in Eq. (8) [19]. Based on the similarity scores, local models are grouped into Similar Models Group $\mathcal{G}_S^{(k)}$ and Non-Similar Models Group $\mathcal{G}_{NS}^{(k)}$.

b) *Global model aggregation:* Two types of global models are created at each IC_k :

- 1) *MLP-SimiFed Global Model (SGM):* The aggregation of similar models is performed using Eq. (11):

$$\theta_{SGM}^{(k)} = \frac{1}{|\mathcal{G}_S^{(k)}|} \sum_{\theta_i \in \mathcal{G}_S^{(k)}} \theta_i \quad (11)$$

- 2) *MLP-Non-SimiFed Global Model (N-SGM):* The aggregation of non-similar models is depicted in Eq. (12):

$$\theta_{N-SGM}^{(k)} = \frac{1}{|\mathcal{G}_{NS}^{(k)}|} \sum_{\theta_i \in \mathcal{G}_{NS}^{(k)}} \theta_i \quad (12)$$

We aggregate both SGM and N-SGM to capture the stability-specificity trade-off: SGM (high-similarity cluster) preserves generalized patterns, while N-SGM retains rare/long-tail and cloud-specific behaviors that are otherwise averaged out; this dual path reduced cross-entropy and Hamming loss compared to single-group aggregation.

B. Second Layer: Unified Global CTI Model

1) *Second-Level Heterogeneous FL Optimization*: The second layer aggregates the global models from Z industry cloud platforms $\{IC_1, IC_2, \dots, IC_Z\}$ to construct the Unified Global CTI Model as demonstrated in Fig. 3. This involves the aggregation of both MLP-SimiFed and MLP-Non-SimiFed models using the weighted averaging method using Eq. (13);

$$\theta_{UG-CTI} = \frac{1}{Z} \sum_{i=1}^Z (\alpha_i \theta_{SGM}^{(i)} + \beta_i \theta_{N-SGM}^{(i)}) \quad (13)$$

where α_i and β_i are the weights assigned to the Similar and Non-Similar models, respectively.

2) *Real-Time Model Updation, Regeneration and Adaptation*: The Unified Global CTI Model is updated in real-time using new threat intelligence data by using Eq. (14):

$$\theta_{UG-CTI}^{(t+1)} = \theta_{UG-CTI}^{(t)} + \mu (\theta_{UG-CTI}^{new} - \theta_{UG-CTI}^{(t)}) \quad (14)$$

where, $\theta_{UG-CTI}^{(t+1)}$ is the updated Unified Global CTI Model, μ is the learning rate for model updates, and θ_{UG-CTI}^{new} is the newly aggregated model based on real-time data. The Unified Global CTI model regeneration is necessary when significant drift or variation in data is detected. This is achieved by recalculating model similarities and performing first-level aggregation using Eq. (15). The final Unified Global Model is then updated by applying Eq. (16):

$$\theta_{newS} = \frac{1}{|G_{newS}|} \sum_{\theta_i \in G_{newS}} \theta_i \quad (15)$$

$$\theta_M^{(2)} = \alpha \cdot \theta_{newS} + (1 - \alpha) \cdot \theta_{newNS} \quad (16)$$

where, θ_{newNS} is the newly aggregated model from non-similar data and α is the weight assigned to the similar model.

C. Third Layer: Meta Unified Global CTI Model

To further enhance the Unified Global CTI Model, transfer learning is applied as mentioned in Eq. (17), where $\theta_{MUG-CTI}$ represents the Meta-Unified Global CTI Model, and $\Delta\theta_{TL}$ denotes the adjustment made through Transfer Learning (TL). The purpose of TL is to adapt the pre-trained Unified Global CTI Model to new environments or specific cloud platforms by leveraging knowledge from the existing model.

$$\theta_{MUG-CTI} = \theta_{UG-CTI} + \Delta\theta_{TL} \quad (17)$$

As depicted in Fig. 3, the transfer learning process begins by selectively unfreezing the outer layers of the trained unified MLP global model, allowing these layers to be fine-tuned to accommodate new data specific to the target domain. The outer layers are often those responsible for higher-level feature

extraction, making them suitable for adaptation to new, unseen environments using Eq. (18):

$$\theta_{TL} = \{W_i, b_i\}_{i=1}^{l_o} \quad \text{for } l_o \in \{l-m, \dots, l\} \quad (18)$$

wherein, W_i and b_i represent the weights and biases of the i -th layer, respectively, and l_o denotes the outer layers of the model that are unfrozen, where m is the number of layers selected for fine-tuning, and l is the total number of layers in the original MLP model. After unfreezing these layers, the model undergoes a fine-tuning process using a domain-specific dataset D_{TL} associated with the new environment. The fine-tuning process can be expressed as an optimization problem, aiming to minimize the loss function $\mathcal{L}_{TL}$ for the adapted model using Eq. (19):

$$\theta_{TL}^* = \arg \min_{\theta_{TL}} \mathcal{L}_{TL}(D_{TL}; \theta_{TL}) \quad (19)$$

The fine-tuned model parameters θ_{TL}^* are then integrated with the existing Unified Global CTI Model parameters θ_{UG-CTI} to form the updated Meta Unified Global CTI Model as stated in Eq. (20):

$$\theta_{MUG-CTI} = \alpha \cdot \theta_{UG-CTI} + (1 - \alpha) \cdot \theta_{TL}^* \quad (20)$$

wherein, α is a weighting factor that balances the contribution of the original model and the fine-tuned model to the final Meta Unified Global CTI Model. This approach ensures that the MUG-CTI Model is robust, proactive, and capable of adapting to a diversity of heterogeneous cyber threats across multiple cloud environments. The adaptability of the model is further enhanced by periodically regenerating and recalculating model similarities, which allows the MUG-CTI Model to remain resilient to shifts in cyber threat landscapes. This adaptability process is expressed as depicted in Eq. (21):

$$\theta_{new}^S = \frac{1}{|G_{new}^S|} \sum_{\theta_{new}^i \in G_{new}^S} \theta_{new}^i \quad (21)$$

The final meta-global model (Eq. 22) is updated by combining the regenerated model θ_{new}^S with the non-shifted model θ_{new}^{NS} as follows:

$$\theta_M^{(2)} = \alpha \cdot \theta_{new}^S + (1 - \alpha) \cdot \theta_{new}^{NS} \quad (22)$$

This ongoing process of adaptation and regeneration ensures that the MUG-CTI Model remains effective in identifying and countering evolving and diverse cyber threats in real-time.

D. Operational Design and Complexity Analysis

Algorithm 1 outlines operational flow of MUG-CTI model. **Time Complexity Analysis**: Step 1: Preprocessing and Input Retrieval is assumed to be completed beforehand, thus this step has $\mathcal{O}(1)$ complexity. Steps 2-3: Training K local models across z communication rounds, with each model having L layers, N neurons per layer, and input size D , involves a complexity of: $\mathcal{O}(K \cdot z \cdot M \cdot L \cdot N \cdot D)$. Step 4: Aggregating SGM and N-SGM models is performed with negligible cost, leading to $\mathcal{O}(1)$ complexity. Step 5: Updating the UGM-CTI model is also performed with constant time complexity of $\mathcal{O}(1)$. Step 6: Transfer learning, which involves

Algorithm 1 MUG-CTI Operational Summary

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1 Input: Pre-processed  $K$  diverse cyber threat databases (CTB);
2 for each Industry Cloud (IC)  $\{IC_1, IC_2, \dots, IC_Z\}$  do
3   Start centralized server and trigger training process for local MLP models;
4   Train local models  $\{LM_1, LM_2, \dots, LM_K\}$  at  $K$  client sites with respective CTB databases;
5   for each communication round  $i \{i_1, i_2, \dots, i_z\}$  do
6     First Layer: Share, aggregate, and update two distinct global models: Similar Group Model (SGM) and Non-Similar Group Model (N-SGM) using Eqs. (9) and (10);
7     Analyze and update global models based on Eqs. (11) and (12);
8 Second Layer: Aggregate all SGM and N-SGM models using Eq. (13) to develop Unified Global Model (UGM-CTI) and update it in real-time using Eqs. (14–16);
9 Third Layer: Unfreeze the outer layers of the UGM-CTI model by applying transfer learning with Eq. (17) to Eq. (22), and train using new cyber threat data to develop the MUG-CTI;
10 Analyze performance of the MUG-CTI model by computing metrics such as cross-entropy, Hamming loss, PR-AUC, and ROC-AUC;

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unfreezing $L - F$ layers and training on a new dataset of size M^* , results in a time complexity of: $\mathcal{O}(M^* \cdot (L - F) \cdot N \cdot D)$. Step 7: Performance evaluation, which computes various metrics on a test set of size M' , results in a complexity of: $\mathcal{O}(M')$. Summing up all steps, the *total time complexity* is: $\mathcal{O}(K \cdot z \cdot M \cdot L \cdot N \cdot D + M^* \cdot (L - F) \cdot N \cdot D + M')$.

E. Security and Robustness Analysis

Federated Learning (FL)-based CTI models are inherently vulnerable to adversarial manipulations such as data poisoning, backdoor attacks, and model update manipulations. In the proposed MUG-CTI model, robustness is formally analyzed by considering the effect of adversarial updates $\mathcal{A}$ on the aggregated global model parameters θ . Let θ_{benign} denote the aggregated model in the absence of adversaries and θ_{adv} the parameters under adversarial influence. The degradation in robustness can be expressed as:

$$\Delta R = \frac{\|\theta_{benign} - \theta_{adv}\|_2}{\|\theta_{benign}\|_2}. \quad (23)$$

A smaller ΔR indicates higher robustness to adversarial manipulation. To mitigate such risks, three mechanisms are incorporated: (i) similarity-based grouping (SGM vs. N-SGM) reduces the impact of poisoned outliers by isolating non-similar updates, (ii) weighted aggregation in the second layer (Eq. 13) limits the influence of malicious participants by controlling α_i, β_i , and (iii) transfer learning with selective layer unfreezing ensures that only higher-level representations adapt, minimizing vulnerability to poisoned low-level features.

Collectively, these mechanisms provide resilience against poisoning and enhance the robustness of the global intelligence fusion process.

F. Interpretability of CTI Insights for Operational Forensics

Beyond predictive accuracy, interpretability is vital for operational forensics. MUG-CTI achieves this through *similarity-driven grouping*, where local models are clustered into generalized ($\mathcal{G}_S$) or specialized ($\mathcal{G}_{NS}$) groups, enabling analysts to trace threat indicators to their behavioral origin. Transfer learning further provides an *explainable threat lineage*, mapping emerging attacks to previously known patterns and clarifying their evolutionary context. These mechanisms not only reduce false positives but also justify alerts and remediation steps, thereby strengthening analyst trust in automated CTI-driven defenses across distributed cloud environments.

III. PERFORMANCE EVALUATION

A. Experimental Set-up and Dataset Utilization

The simulation experiments are conducted on a server machine assembled with two Intel® Xeon® Silver 4114 CPUs with a 40 core processor and a 2.20 GHz clock speed. The computation machine is deployed with 64-bit Ubuntu 16.04 LTS, having 128 GB RAM. For performance evaluation, we developed a test-bed using the Python Flower framework, enabling federated client-server interactions. In this setting, each client (simulated container or VM) trains a local model using its respective cyber threat database. The datasets employed in this study are summarized in Table II, covering four distinct categories from different industry domains:

- *NetFlow*: NF-UNSW-NB15 [20] with 1.623×10^6 samples and 43 features.
- *VM Threat*: Ten VM Threat databases, each with 15,000 samples and 11 features.
- *Intrusion*: CICIDS2017 [21] with 2,830,743 samples and 80 features.
- *Authentication*: Ten Access databases [22], each with 11,128 samples and 8 features.

The training process was conducted in three hierarchical layers:

- 1) *Layer 1 (Local Model Training)*: Each dataset was treated as a distinct cyber threat database representing an industry-specific cloud. At this stage, multiple local models were trained per industry cloud. These local models were grouped into *similar* and *non-similar* categories, and used to build two corresponding global models for each industry cloud.
- 2) *Layer 2 (Unified Global Model)*: The similar and non-similar global models from all industry clouds were then considered as local inputs to a higher-level aggregation process. This aggregation produced a *Unified Global Model*, representing consolidated knowledge across all clouds.
- 3) *Layer 3 (Transfer Learning Augmentation)*: The unified global model was further enhanced via Transfer Learning using newly available cyber threat data traces. This

TABLE II
CHARACTERISTICS OF EVALUATED CYBER THREAT DATASETS

Category	Dataset (#)	Count	Mean (%)	St.dev	Features
NetFlow	NF-UNSW-NB15 [20]	1.623×10^6	4.46×10^{-2}	2.06×10^{-1}	43
VM Threat	VM Threat DB (10)	15000	0.5065	0.4999	11
Intrusion	CICIDS2017 [21]	2,830,743	0.1506	0.4423	80
Authentication	Access DB (10) [22]	11128	0.050863	0.21972	8

TABLE III
EXPERIMENTAL SET-UP PARAMETERS AND VALUES

Parameter	Value/Description
Number of Industry Clouds (K)	3-25
Number of local models (N_k)	varies from 2 to 10 per IC
Local Model	MLP with 3 hidden layers
Input Layer Sizes	64 neurons
Hidden Layer Sizes	32 neurons
Output Layer Sizes	number of classes
Activation Function	ReLU
Loss Function (L_k)	Cross-Entropy Loss
Optimizer	Adam Optimizer
Learning Rate (η)	0.001 (tuned between 0.0001 and 0.01)
Regularization Parameter (λ)	0.01 (tuned between 0.001 and 0.1)
First-Level FL Aggregation (SGM and N-SGM)	Weighted Average based on dataset size
Second-Level FL Aggregation (UG-CTI)	Equal Weighted Average of SGM and N-SGM
Transfer Learning Regularization (γ)	0.1 (tuned between 0.01 and 1)
Real-Time Update Momentum Coefficient (μ)	0.8 (tuned between 0.5 and 0.9)
Number of Global Communication Rounds	50 (varies from 20 to 100)
Similarity threshold ϵ	0.6
Batch Size	32 (tuned between 16 and 128)
Number of Epochs per Local Training	10 (varies from 5 to 20)
Transfer Learning Data Size	1000 samples (varies from 500 to 2000)
Evaluation Metric	Accuracy, F1-Score, Recall, Hamming loss
Software Framework	TensorFlow/PyTorch and Python Flower set-up

resulted in the final *Meta-Unified Global Cyber threat Intelligence (MUG-CTI)* model.

This layered training approach allowed us to integrate heterogeneous datasets holistically. Unlike conventional studies that evaluate performance on a per-dataset basis, our methodology leverages all datasets simultaneously to investigate the performance of the complete MUG-CTI model in a unified manner. The values of parameters used during the experimental study with the details are listed in Table III.

B. Results and Discussion

Table IV presents the key performance metrics of the proposed MUG-CTI model across varying numbers of cross-clouds (IC #), combinations of local models (LM #), and communication rounds (CR #). The table reports accuracy (A %), cross-entropy error, F1 score, recall (RC), precision (PR), Hamming loss, training time (TT), and memory consumption (MC %). These metrics not only validate predictive performance but also implicitly capture critical aspects of federated learning in inter-cloud environments, including communication cost, latency, and scalability under load. MUG-CTI demonstrates robust and consistent performance as IC #, LM #, and CR # increase, confirming its ability to scale across diverse collaborative networks while keeping communication overhead and latency growth minimal. Transfer learning further enhances its ability to generalize, enabling efficient detection of unseen cyberattacks. The model achieves a high accuracy range [96.79%–99.79%], with cross-entropy as low as [0.0017–0.0099], and excellent F1, recall, and precision scores of up to 0.9951, 0.9983, and 0.9968, respectively. Hamming loss, which measures prediction errors in multi-class threat classification, remains impressively low [0.0026–0.0096].

Importantly, training time and memory consumption increase only marginally with larger IC #, LM #, and CR #, reflecting efficient handling of communication and computational loads during FL training. This demonstrates that MUG-CTI not only delivers high predictive accuracy but also maintains low communication cost, minimal latency, and strong scalability under load, making it well-suited for real-world inter-cloud cyber threat intelligence systems.

Fig. 4 (a–c) present the AUC-ROC curves for 5, 11, and 21 ICs, respectively, showing the trade-off between the true positive rate (TPR) and false positive rate (FPR) across nine threat classes. A clear divergence in performance is observed: classes such as C0, C1, C7, and C9 consistently achieve AUC values close to 1.0, reflecting highly reliable detection with minimal false alarms. In contrast, classes C3, C4, C5, C6, and C8 remain around 0.50, which is equivalent to random guessing. This disparity arises primarily from severe class imbalance, where underrepresented threats lack sufficient training examples, and from overlapping feature distributions that reduce the separability of certain malicious behaviors from benign traffic.

Complementing this analysis, the PR-AUC curves in Fig. 5 (a–c) provide further insights under the imbalanced nature of cyber threat datasets, where the proportion of positive (threat) samples is much smaller than negative (benign) instances. While ROC curves may overestimate performance under imbalance, PR curves highlight the true discriminative capability by emphasizing precision (false positive reduction) alongside recall (detection coverage). The results demonstrate that high-performing classes (C0, C1, C7, C9) maintain strong PR-AUC values above 0.60, whereas, low-performing classes (C3, C5) struggle to exceed 0.10–0.20, confirming that poor recall and high false positives dominate their detection. Notably, as the number of ICs increases from 5 to 21, most classes exhibit improved stability in both ROC and PR spaces, suggesting that richer contextual information aids in mitigating variance. However, the persistent underperformance of minority or overlapping classes underscores the need for advanced countermeasures such as class re-weighting, synthetic oversampling, or disentangled feature learning to strengthen model robustness across all threat scenarios.

C. Ablation Analysis

Table V presents the ablation study, highlighting the effectiveness of the proposed MUG-CTI model compared to state-of-the-art baselines (FL, FL+Sim, FL+TL, and FL+Sim+TL) across multiple cross-cloud settings (3, 5, 11, and 21). Results show that MUG-CTI consistently outperforms competing models, achieving the highest precision, recall, and F1 scores (up to 1.00), and delivering up to 7.52% performance improvement in some scenarios.

The superior performance of MUG-CTI arises from the synergistic integration of federated learning, cosine similarity, non-similarity integration, and transfer learning, which collectively enhance adaptability, minimize classification errors, and ensure robust threat detection in highly dynamic environments. Despite its algorithmic sophistication, MUG-CTI sustains

TABLE IV
PERFORMANCE METRICS FOR MUG-CTI-BASED INDUSTRIAL COLLABORATIVE CLOUD NETWORKS

IC #	LM #	CR #	A (%)	Entropy	F1 Score	RC	PR	Ham. Loss	TT (msec)	MC (%)
3	5	10	98.97	0.0091	0.9891	0.9592	0.9689	0.0096	32.85	63.80
		20	98.01	0.0087	0.9971	0.9943	0.9834	0.0081	34.11	62.97
	10	10	97.13	0.0017	0.9869	0.9869	0.9902	0.0096	32.76	67.76
		20	98.01	0.0097	0.9951	0.9983	0.9764	0.0074	35.56	66.59
5	5	10	97.02	0.0083	0.9675	0.9834	0.9799	0.0071	31.97	64.11
		20	98.63	0.0061	0.9883	0.9711	0.9745	0.0077	37.67	68.12
	10	10	96.79	0.0051	0.9991	0.9981	0.9902	0.0088	36.02	68.14
		20	99.79	0.0043	0.9771	0.9886	0.9877	0.0064	32.76	63.77
11	5	10	98.14	0.0071	0.9777	0.9789	0.9596	0.0066	32.37	63.40
		20	99.73	0.0047	0.9903	0.9902	0.9834	0.0048	38.16	67.56
	10	10	96.95	0.0044	0.9782	0.9821	0.9802	0.0038	35.22	69.89
		20	99.79	0.0043	0.9701	0.9992	0.9968	0.0054	37.14	62.94
21	5	10	96.55	0.0058	0.9862	0.9889	0.9677	0.0087	33.76	62.97
		20	98.86	0.0032	0.9711	0.9892	0.9845	0.0031	39.10	65.42
	10	10	98.85	0.0036	0.9893	0.9724	0.9905	0.0026	37.12	67.02
		20	98.99	0.0024	0.9891	0.9986	0.9878	0.0078	38.59	68.04

IC#: number of industry cross-clouds; LM#: number of local models; CR #: number of communication rounds; A (%): Accuracy of threat prediction; RC: recall values; PR: precision; Ham. loss: hamming loss; TT: training time; MC: memory consumption

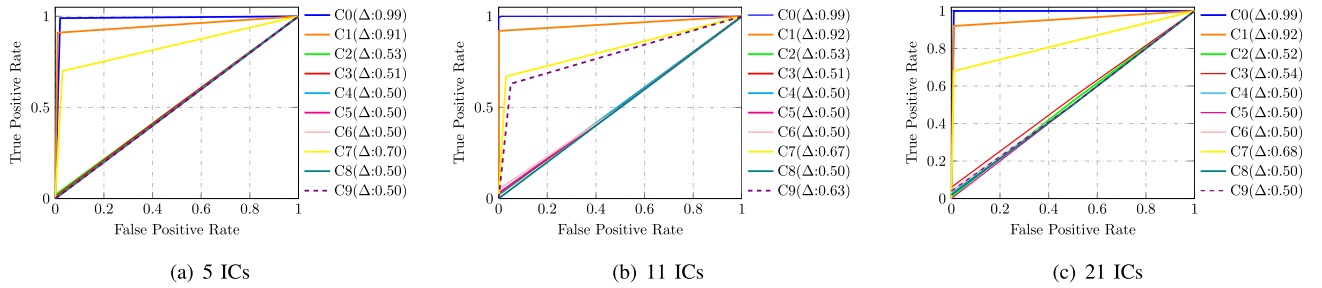


Fig. 4. AUC-ROC curves.

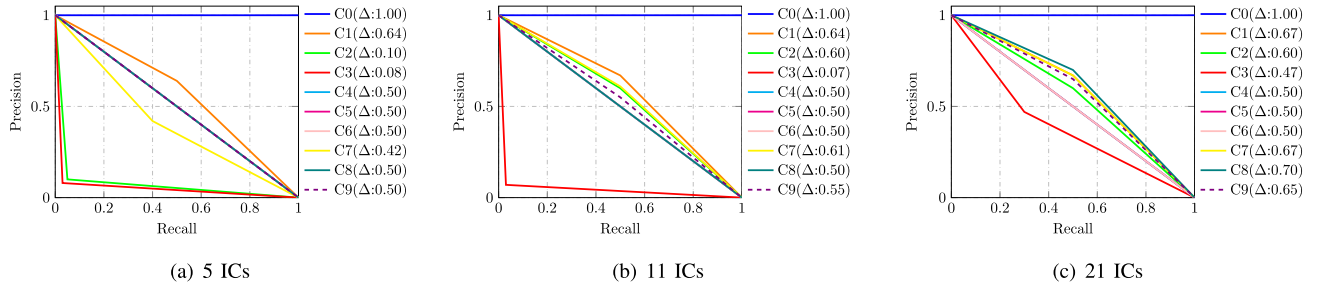


Fig. 5. Precision-recall curves.

competitive memory efficiency (around 69%) and scalable training performance, underscoring its practicality for real-world deployment. Most notably, it achieves an exceptionally low Hamming Loss of 0.0028, validating its reliability for multi-class cyber threat classification under evolving attack patterns. While MUG-CTI incurs a marginal increase in training time compared to baseline models, this limitation is manageable and improvable through strategies such as parallelized training, lightweight architectural variants, and adaptive fine-tuning. These refinements can further strengthen scalability without sacrificing detection accuracy, providing a balanced, transparent, and forward-looking evaluation of the proposed methodology. The ablation results underline three key strengths: (i) effective knowledge transfer across heterogeneous clouds, (ii) minimized false positives/negatives

through similarity optimization, and (iii) resilience in large-scale deployments. These features make MUG-CTI highly suitable for real-time cyber threat intelligence in complex multi-cloud infrastructures.

D. Comparison

The key performance indicators (KPIs) of MUG-CTI are rigorously evaluated against baseline methods, including *Federated Learning* (FL+Avg) [11], *Federated Learning + Cosine Similarity* (FL+Sim), *Federated Learning + Transfer Learning* (FL+TL), and the proposed variant *Federated Learning + Cosine Similarity + Transfer Learning* (FL+Sim+TL) across multiple cross-cloud environments in Section III-D.1. Additionally, Section III-D.2 presents the performance of MUG-CTI is benchmarked against *Support Vector Machine*

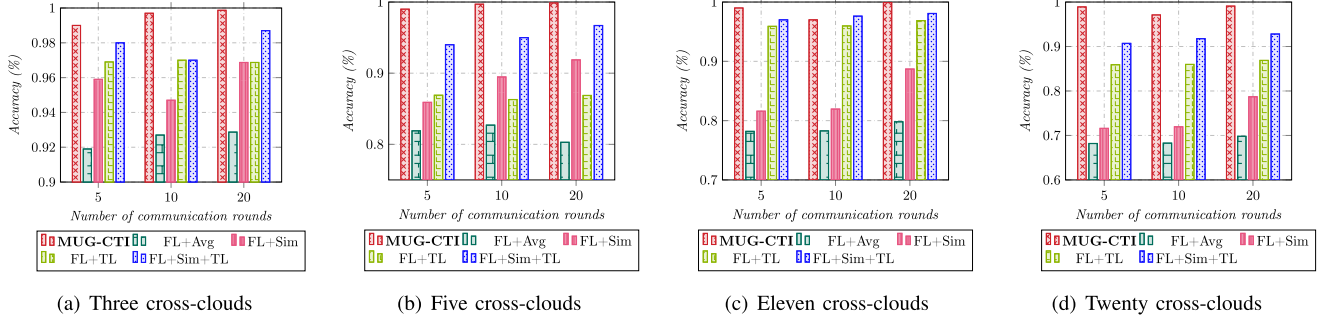


Fig. 6. Comparative accuracy for multi-threat classification in cross-clouds.

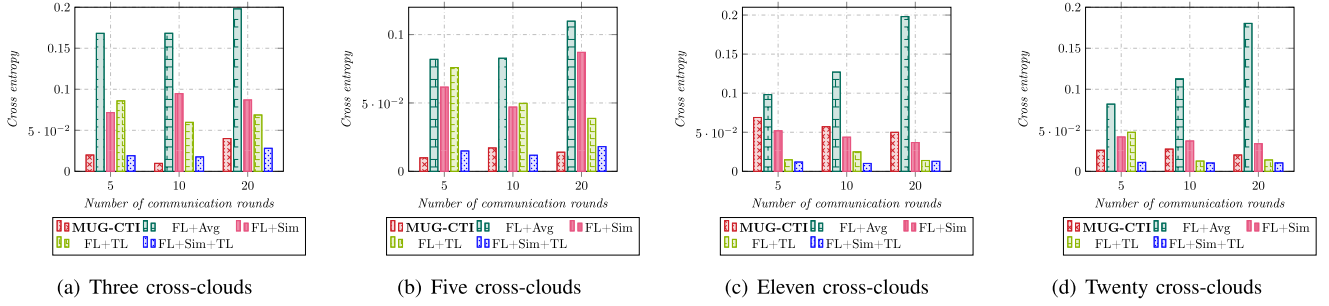


Fig. 7. Comparative entropy for multi-threat classification in cross-clouds.

TABLE V
ABLATION STUDY: KPI ANALYSIS OF THE MUG-CTI MODEL

IC #	Model	PR	RC	F1	MC (%)	TT (ms)	HL
3	FL	0.97	0.96	0.97	68.08	22.08	0.0198
	FL+Sim	0.98	0.97	0.93	62.89	28.95	0.0793
	FL+TL	0.97	0.96	0.94	60.97	29.97	0.0729
	FL+Sim+TL	0.98	0.98	0.98	59.96	20.96	0.0181
	MUG-CTI	0.99	0.99	0.99	63.85	32.85	0.0096
5	FL	0.97	0.97	0.95	71.08	21.56	0.0319
	FL+Sim	0.98	0.95	0.93	66.09	25.77	0.0821
	FL+TL	0.97	0.93	0.94	65.12	29.07	0.0541
	FL+Sim+TL	0.99	0.96	0.99	64.04	27.23	0.0221
	MUG-CTI	1.00	0.99	0.99	70.58	39.01	0.0071
11	FL	0.98	0.98	0.96	70.34	24.67	0.0197
	FL+Sim	0.98	0.95	0.93	63.81	26.89	0.0181
	FL+TL	0.96	0.95	0.96	68.45	29.07	0.0431
	FL+Sim+TL	0.99	0.99	0.99	68.67	29.93	0.0092
	MUG-CTI	1.00	1.00	1.00	69.58	42.01	0.0033
21	FL	0.96	0.94	0.96	69.94	28.07	0.0107
	FL+Sim	0.97	0.97	0.98	64.01	29.93	0.0112
	FL+TL	0.97	0.96	0.97	67.75	34.07	0.0381
	FL+Sim+TL	0.99	0.98	0.98	69.93	49.93	0.0090
	MUG-CTI	1.00	1.00	0.99	69.23	67.11	0.0028

IC#: Industry cross-clouds; PR: Precision; RC: Recall; F1: F1-score; MC: Memory consumption; TT: Training time; HL: Hamming Loss.

(SVM) [14], *Multi-Layer Perceptron* (MLP) [17], *Random Forest* (RF) [23], FL+Avg [11], FL+Sim, FL+TL, and FL+Sim+TL in a single-cloud environment.

1) *For Cross-Clouds Environment*: Fig. 6 compares the accuracy of threat prediction across three (Fig. 6a), five (Fig. 6b), eleven (Fig. 6c), and twenty-one (Fig. 6d) cross-cloud environments. The comparative Federated learning models including FL+Avg, FL+Sim, FL+TL, FL+Sim+TL, are evaluated over 5, 10, and 20 communication rounds. MUG-CTI consistently outperforms these models, achieving accuracy

improvements of 20.70%, 7.60%, 15.12%, and 3.12% over FL+Avg, FL+Sim, FL+TL, and FL+Sim+TL, respectively. Additionally, Fig. 7 reveals that MUG-CTI exhibits the lowest cross-entropy error, reducing it by 98.33%, 93.33%, 75.12%, and 66.67% compared to the baseline methods with highest reduction in hamming loss up to 73.8% against FL+Avg. These results are attributed to superior integration of cosine similarity, distinct non-similarity, and transfer learning of MUG-CTI model, enabling efficient knowledge transfer across diverse clouds and improving threat prediction accuracy while minimizing errors.

2) *For Single Cloud Environment*: The MUG-CTI model and its federated learning-driven variants are compared against traditional machine learning-based CTI models, such as SVM, Random Forest (RF), and Multi-Layer Perceptron (MLP), in single cloud environments. Fig. 8 illustrates the performance across key KPIs, where traditional models are trained using a centralized approach, while federated learning-driven models utilize dataset fragmentation into ten parts for local and global model training, incorporating transfer learning as necessary. Results demonstrate that MUG-CTI outperforms traditional models with up to 99.98% accuracy in both training and testing phases (Fig. 8a). It also achieves the highest precision (Fig. 8c), recall (Fig. 8e), and F1 score (Fig. 8f) within the [0.97-0.99] range, maintaining consistent performance, unlike the traditional models, whose metrics drop significantly during testing. Furthermore, MUG-CTI shows 43.06% more accuracy than SVM, with lowest cross-entropy and Hamming loss values, outperforming SVM by 98.33% and 93.33%, respectively, while maintaining comparable resource consumption, with 24.40% lower memory usage and 45.45% more training time than SVM. The federated learning approach of MUG-

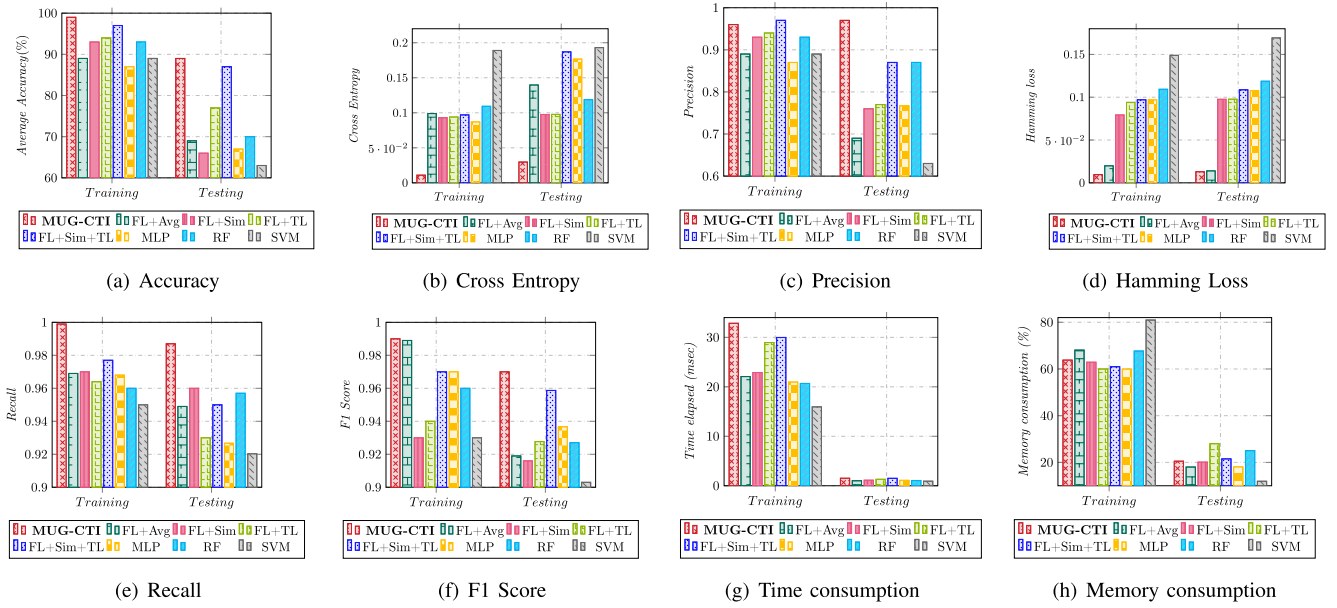


Fig. 8. Performance metrics comparison for multi-threat classification across single cloud environment.

CTI enables efficient handling of multi-class and multi-label threat classification across distributed datasets in cross-cloud environments, ensuring high classification accuracy and adaptability. While MUG-CTI requires slightly longer training compared to other federated models (FL+Sim+TL), this overhead can be mitigated through parallelized training, optimized resource allocation, and incremental learning strategies. In return, the model demonstrates exceptional adaptability to diverse and complex cyber-attacks across cross-cloud infrastructures, ensuring consistent reliability even under dynamic threat landscapes. Its scalability, strengthened by transfer learning, allows seamless integration into heterogeneous multi-cloud environments without significant reconfiguration. By balancing modest training costs with substantial gains in accuracy, precision, recall, and resilience, MUG-CTI provides a durable and future-proof cybersecurity solution capable of sustaining performance at scale.

IV. CONCLUSION AND FUTURE WORK

The proposed multi-diverse CTI model substantially enhances the security of industrial cross-cloud networks by addressing key challenges of scalability, heterogeneity, and real-time adaptability. Its three-layered meta-unified framework, combining Multi-Layer Perceptron (MLP)-based Federated Learning, unified global aggregation, and Transfer Learning, provides a comprehensive strategy for detecting and mitigating cyber threats. This architecture not only consolidates intelligence across heterogeneous cloud platforms but also continuously updates in real time, promoting resilient and adaptive defenses in dynamic threat landscapes. The MLP-based design serves as the core computational backbone, chosen for its efficiency, scalability, and compatibility with federated learning scenarios where lightweight models are advantageous. While highly effective, reliance on MLPs may constrain the capacity to capture complex relational

structures and temporal patterns inherent in CTI datasets. More expressive architectures, such as Graph Neural Networks or transformers, could offer enhanced modeling of interdependencies among threat entities and sequential dynamics. Nevertheless, the current MLP-based approach strikes a careful balance between accuracy, interpretability, and resource efficiency, establishing a strong foundation for future exploration of these advanced architectures in cross-cloud CTI analytics.

Future work will focus on further enhancing adaptability to emerging threats by integrating explainable AI techniques for transparent and interpretable CTI outputs, extending evaluations to real-world industrial multi-cloud deployments, and performing robustness testing against adversarial attacks in federated settings. Additionally, expanded test-bed simulations across diverse industrial scenarios will provide rigorous assessment of scalability and practical applicability, ensuring the framework's reliability, trustworthiness, and readiness for deployment in large-scale heterogeneous cloud ecosystems.

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